



Location API 1.0

Developer Guide

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About this guide

This guide is designed to help you access location metadata for your devices, using the Location API.

Extra reading

For general information, see the online help: <https://docs.eseye.com/Content/Home.htm>

For information about alternate ways to locate your device, see [Locating devices using a cellular connection](#).

You can also manage SIM cards through the web portal. For more information, see the [8295 SIAM User Guide \(PDF\)](#).

For AWS-integrated devices and viewing device location using the shadow, see [Managing the device shadow](#).

Standards and conventions

This guide uses consistent visual cues and standard text styles to help you locate and interpret information easily.

Style	Description
 Note	Extra information or a recommendation related to the current topic.
 Tip	Good-to-know information that helps users complete a procedure or understand a topic.
 Warning	Information that alerts the user about significant or critical actions or outcomes.
<i>Title names</i>	Window or section names, denoting a title, appear in italics.
Field or button name	Element names in a user interface, for example fields and buttons, appear in bold.
Ctrl+X; Ctrl+click	A key combination with a plus sign separating two key names or a key name and a mouse action, indicates that you hold down the first key while pressing the second key or performing the mouse action.
Cross reference <i>Title</i> and page	Cross references appear in italics, for example: For more information, see <i>Copyright</i> on page 2. Select the cross reference to view it.
<u>Hyperlinks</u>	Underlined cross references are hyperlinks to electronic forms of the document. Select the hyperlink to open the cross reference.
AT Commands	
Commands	Command formats are displayed in monospaced typeface.
<Parameter>	Angle brackets enclose the AT Command parameter, for example <topic>. The brackets do not appear in the command line.
"ParameterString"	Quotation marks enclose parameter strings.
<i>CommandValue</i>	Italics in a command depicts values or examples that need replacing with your specific parameters.
[CommandOptionalEntry]	Square brackets display optional entries.
ATCommandResponse	Returned responses to AT Commands are displayed in monospaced bold typeface.
<ASCIICHARACTERS>	Returned ASCII characters are in uppercase.

Location API introduction

The Location API returns detailed SIM location information, such as latitude, longitude, and cell mast information. You can also use it to fetch a compressed Keyhole Markup Language (KMZ) file for use with Google Earth. You can only view information for SIMs that are already linked to your SIAM account.

Eseye makes data available for two months. You must store any historic data you want to retain.

Before you begin

You will need:

- A SIAM account, with all your SIM cards linked to this account.
- Your SIAM account portfolioID. Contact Eseye Support for more information: support@eseye.com.
- A static IP address that Eseye will whitelist to access the Location API.
- Contact your Account Manager to set up Location API.

Accessing the API

Use the following URL to access the latest version of the Tigrillo API:

- `https://location.eseye.com/index/<method>`
where `<method>` is the method used, for example `locatedevice`.

Location API requests

You can call the Location API in any API development app using the following commands:

- `locatedevice` – see *POST /index/locatedevice* on page 5
- `locatemast` – see *POST /index/locatemast* on page 7
- `locatecalculate` – see *POST /index/locatecalculate* on page 10
- `kmz` – see *POST /index/kmz* on page 13

All information is sent in JSON.

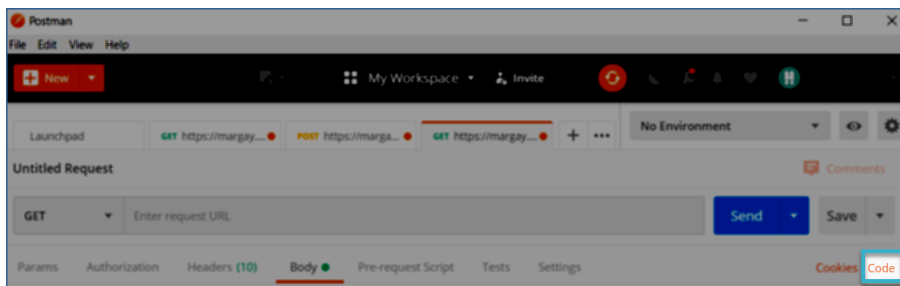


Use plain text to send requests.



All Eseye API examples are written for Postman. For information about using Postman, see the [8314 Postman User Guide \(PDF\)](#).

Using Postman, if required, you can select **Code** to generate code snippets.



For more information, see *Troubleshooting errors* on page 15.

Configuring headers

Headers accompany the body of a request, containing information about the request and how to process it. All Location API requests use the same headers.

Headers



If you are using Postman, the following headers are automatically generated.

Schema

Object name	Default value (Postman)	Description	Required?
Cookie	PHPSESSID=<CookieNumber>	Sends cookies that are associated with this endpoint.	✓
Cache-Control	no-cache	We recommend using this header. Prevents the server returning a stale response when you make repeated requests.	
Postman-Token	<calculated when request is sent>	If you are using Postman, we recommend this header, which appends a random UUID to every outgoing request. This aids debugging, by ensuring separate requests are distinct in the receiving server.	
Content-Type	application/json	Helps the server identify the media type of the current request body.	✓
Content-Length	<calculated when request is sent>	For parsing purposes. Indicates the size of the current request body to the server.	
Host	<calculated when request is sent>	Helps the server identify the domain name where the request is sent.	✓
User-Agent	PostmanRuntime/<PostmanVersion>	If you are using Postman, we recommend using this header. Helps the server identify Postman as the HTTP requesting application or client.	
Accept	*/*	If you are using Postman, we recommend using this header. The "Accept: */*" header tells the server that Postman can understand and process all forms of response content types.	

Object name	Default value (Postman)	Description	Required?
Accept-Encoding	gzip, deflate, br	If you are using Postman, we recommend using this header. Indicates to the server that the Postman HTTP client supports a defined list of content-encoding or compression algorithms as response.	
Connection	keep-alive	We recommend using this header. Indicates to the server to keep the underlying network connection open after receiving the current response. This enables you to reuse the same connection for faster response times in subsequent requests to the same server.	

Example

Headers	
1	'headers': {
2	'Content-Type': 'application/json',
3	'Cookie': 'PHPSESSID=3cp0ct8ofutokn6q950qpg5555'
4	},

POST /index/locatedevice

Interrogates the network for current location information. If the device is offline or the current location is not available, returns the last recorded value.

Authentication

- User required to login

Headers

For information about headers, see *Configuring headers* on page 3.

Request Body application/json

Schema



Use null without quotes as the input if you do not wish to use these objects. The value will remain as previously set. If you want to change the object value to a new value, replace null with "<value>" within quotes, where <value> is the required value. To remove an existing value, type an empty string. For example: "friendlyName" : "" ,

JSON key	Description	Required?
request_id	A request identifier for tracking responses.	
iccid	The unique Integrated Circuit Card Identifier of the SIM you want to locate. Use either this parameter or msisdn to identify the SIM.	✓ or msisdn
msisdn	The unique primary SIM phone number (Mobile Station International ISDN Number) of the SIM you want to locate. Use either this parameter or iccid to identify the SIM.	✓ or iccid
username	The Eseye-assigned Location API username	✓
password	The Eseye-assigned Location API password	✓

Example

Body raw JSON

```
{
  "request_id" : "42",
  "iccid" : "8944538523007980017" ,
  "username" : "ExampleUsername" ,
  "password" : "ExamplePassword" ,
}
```

Responses application/json

Schema

JSON key	Description
iccid	The unique Integrated Circuit Card Identifier for the located SIM.
msisdn	The unique primary SIM phone number (Mobile Station International ISDN Number).
lat	Latitude value of the last cell mast that the SIM used.
lon	Longitude value of the last cell mast that the SIM used.
countryname	The last recorded country the SIM was in.
countrycode	The ISO 3166 Alpha-2 code of the last recorded country the SIM was in. For more information, see the ISO Online Browsing Platform (ODP) Country Codes .
timezone	The TZ database name of the last recorded timezone the SIM was in. For more information, see the List of TZ database time zones .
status	The array, containing details about the request status.
status	"OK" – the request processed successfully "ERR" – the request failed
errorCode	" " – the request processed successfully "<ErrorCodeNumber>" – the request failed because of this specific error. For more information, see <i>Troubleshooting errors</i> on page 15.
errorMessage	" " – the request processed successfully "<ErrorCodeMessage>" – displays an explanation of the error if the request failed. For more information, see <i>Troubleshooting errors</i> on page 15.
request_id	The returned request identifier, for tracking responses.

Example

JSON
<pre>{ "iccid": "8944538523007980017", "msisdn": "447924167590", "lat": 51.2394, "lon": -0.6036, "countryname": "United Kingdom", "countrycode": "GB", "timezone": "Europe/London", "status": { "status": "OK", "errorCode": "", "errorMessage": "" }, "request_id": "42" }</pre>

POST /index/locatemast

Interrogates the network for the latitude, longitude, country name, country code, operator name and time zone of a specific cell tower. If the tower is unavailable, returns the last recorded values for that tower.

Authentication

- User required to login

Headers

For information about headers, see *Configuring headers* on page 3.

Request Body application/json

Schema



Use null without quotes as the input if you do not wish to use these objects. The value will remain as previously set. If you want to change the object value to a new value, replace null with "<value>" within quotes, where <value> is the required value. To remove an existing value, type an empty string. For example: "friendlyName" : "" ,

JSON key	Description	Required?
request_id	A request identifier for tracking responses.	
username	The Eseye-assigned Location API username	✓
password	The Eseye-assigned Location API password	✓
mast	The array, containing details about the specific cell tower.	
mcc	The Mobile Country Code – identifies the country where the cell tower exists, in three-digit numeric format. For more information, see Mobile country codes .	✓
mnc	The Mobile Network Code – identifies the network operator of the network that uses the cell tower, in two-digit numeric format. For more information, see: Mobile network codes listed by country .	✓
lac	The Location Area Code of the cell tower, in decimal format. Tracks the physical location of the SIM within the network. Use either this parameter or LAI to identify the cell tower.	✓ or LAI
lai	The Location Area Identity of the cell tower, in hexadecimal format. Tracks the physical location of the SIM within the network. Use either this parameter or LAC to identify the cell tower.	✓ or LAC
cellid	Cell Identification, in hexadecimal format. Identifies the cell tower in a location area. Use this parameter, or CI.	✓ or CI
ci	The Cell Identification, in hexadecimal format. Identifies the cell tower in a location area. Use either this parameter, or cellid.	✓ or cellid

Example

Body raw JSON

```
{
  "request_id" : "1" ,
  "username" : "ExampleUsername" ,
  "password" : "ExamplePassword " ,
  "mast" : {
    "mcc" : "234" ,
    "mnc" : "30" ,
    "lai" : "47A" ,
    "ci" : "c0f6",
    "lac" : "1234" ,
    "cellid" : "4567"
  }
}
```

Responses application/json

Schema

JSON key	Description
status	The array, containing details about the request status.
errorCode	"" – the request processed successfully "<ErrorCodeNumber>" – the request failed because of this specific error. For more information, see <i>Troubleshooting errors</i> on page 15.
errorMessage	"" – the request processed successfully "<ErrorCodeMessage>" – displays an explanation of the error if the request failed. For more information, see <i>Troubleshooting errors</i> on page 15.
status	"OK" – the request processed successfully "ERR" – the request failed
request_id	The returned request identifier, for tracking responses.
mast	The array, containing details about the cell tower.
mcc	The Mobile Country Code – identifies the country where the cell tower exists, in three-digit numeric format. For more information, see Mobile country codes .
mnc	The Mobile Network Code – identifies the network operator of the network that uses the cell tower, in two-digit numeric format. For more information, see: Mobile network codes listed by country .
lai	Location Area Identity of the cell tower, in hexadecimal format. Tracks the physical location of the SIM within the network.
ci	The Cell Identification, in hexadecimal format. Identifies the cell tower in a location area.
lat	Cell tower latitude value.
lon	Cell tower longitude value.
countryname	The name of the country the cell tower is in.
countrycode	The country ISO 3166 Alpha-2 code. For more information, see the ISO Online

JSON key	Description
	Browsing Platform (ODP) Country Codes.
operatorname	The name of the network operator using the cell tower.
timezone	The TZ database name of the timezone where the cell tower exists. For more information, see the List of TZ database time zones.

Example

JSON
<pre>{ "status": { "status": "OK", "errorCode": "", "errorMessage": "" }, "request_id": "23" "mast": { "mcc": "234", "mnc": "30", "lai": "47A" "ci": "c0f6cf", "lat": 51.245, "lon": -0.6067, "countryname": "United Kingdom", "countrycode": "GB", "operatorname": "EE UK", "timezone": "Europe/London", } }</pre>

POST /index/locatecalculate

Returns an estimated device location radius, based upon information from three or more cell towers.

Authentication

- User required to login

Headers

For information about headers, see *Configuring headers* on page 3.

Request Body application/json

Schema



Use null without quotes as the input if you do not wish to use these objects. The value will remain as previously set. If you want to change the object value to a new value, replace null with "<value>" within quotes, where <value> is the required value. To remove an existing value, type an empty string. For example: "friendlyName" : "" ,

JSON key	Description	Required?
request_id	A request identifier for tracking responses.	
iccid	The unique Integrated Circuit Card Identifier of the SIM you want to locate. Use either this parameter or msisdn to identify the SIM.	✓ or msisdn
msisdn	The unique primary SIM phone number (Mobile Station International ISDN Number) of the SIM you want to locate. Use either this parameter or iccid to identify the SIM.	✓ or iccid
username	The Eseye-assigned Location API username	✓
password	The Eseye-assigned Location API password	✓
mast	The array, containing details about the specific cell tower.	
mcc	The Mobile Country Code – identifies the country where the cell tower exists, in three-digit numeric format. For more information, see Mobile country codes .	✓
mnc	The Mobile Network Code – identifies the network operator of the network that uses the cell tower, in two-digit numeric format. For more information, see: Mobile network codes listed by country .	✓
lac	The Location Area Code of the cell tower, in decimal format. Tracks the physical location of the SIM within the network. Use either this parameter or LAI to identify the cell tower.	✓ or LAI
lai	The Location Area Identity of the cell tower, in hexadecimal format. Tracks the physical location of the SIM within the network. Use either this parameter or LAC to identify the cell tower.	✓ or LAC
cellid	Cell Identification, in hexadecimal format. Identifies the cell tower in a location	✓ or CI

JSON key	Description	Required?
	area. Use either this parameter, or <code>CI</code> .	
<code>ci</code>	The Cell Identification, in hexadecimal format. Identifies the cell tower in a location area. Use either this parameter, or <code>cellid</code> .	✔ or <code>cellid</code>
<code>ta</code>	The timing advance value returned by the modem.	
<code>power</code>	The Received Signal Strength Indicator (RSSI) value, which represents the power of a received signal on a device. The higher the RSSI value, the stronger the signal.	

Example

Body raw JSON
<pre>{ "request_id" : "1", "ICCID" : "8944122650264416971", "username" : "ExampleUsername" , "password" : "ExamplePassword" , "mast":[{ "mcc":"234", "mnc":"10", "lac":"13182", "cellid":"30151", "ta":"2", "power":"79" }, { "mcc":"234", "mnc":"15", "lac":"179", "cellid":"25009", "ta":"5", "power":"93" }, { "mcc":"234", "mnc":"33", "lac":"638", "cellid":"1769", "ta":"2", "power":"88" }] }</pre>

Responses application/json

Schema

JSON key	Description
<code>status</code>	The array, containing details about the request status.
<code>errorCode</code>	"" – the request processed successfully "<ErrorCodeNumber>" – the request failed because of this specific error. For more information, see <i>Troubleshooting errors</i> on page 15.

JSON key	Description
errorMessage	" " – the request processed successfully "<ErrorCodeMessage>" – displays an explanation of the error if the request failed. For more information, see <i>Troubleshooting errors</i> on page 15.
status	"OK" – the request processed successfully "ERR" – the request failed
request_id	The returned request identifier, for tracking responses.
mast	The array, containing details about the cell tower.
lat	Cell tower latitude value.
lon	Cell tower longitude value.
radius	The estimated device location area radius, in kilometres.
confidence	The certainty that a device is in the given radius. Measured on a scale from 0 to 1, where 1 is certain.
countryname	The name of the country the cell tower is in.
countrycode	The country ISO 3166 Alpha-2 code. For more information, see the ISO Online Browsing Platform (ODP) Country Codes .
timezone	The TZ database name of the timezone where the cell tower exists. For more information, see the List of TZ database time zones .

Example

JSON
<pre>{ "status": { "status": "OK", "errorCode": "", "errorMessage": "" }, "request_id": "1", "mast": { "lat": 51.2275, "lon": -0.639, "radius": 3.23, "confidence": 0.32, "countryname": "United Kingdom", "countrycode": "GB", "timezone": "Europe/London" } }</pre>

POST /index/kmz

Returns a compressed Keyhole Markup Language file detailing the Google Earth location of one or more SIMs. You will need to install Google Earth or Google My Maps in order to import the file and view the mapped locations.

kmz – Keyhole Markup Language Zipped

Authentication

- User required to login

Headers

For information about headers, see *Configuring headers* on page 3.

Request Body application/json

Schema



Use null without quotes as the input if you do not wish to use these objects. The value will remain as previously set. If you want to change the object value to a new value, replace null with "<value>" within quotes, where <value> is the required value. To remove an existing value, type an empty string. For example: "friendlyName" : "" ,

JSON key	Description	Required?
request_id	A request identifier for tracking responses.	
username	The Eseye-assigned Location API username	✓
password	The Eseye-assigned Location API password	✓
group	The group name for the SIMs. If left blank, the request is not limited to group.	
ICCID	The array, containing the list of one or more SIMs that you want to locate using their Integrated Circuit Card IDs (ICCID).	
ICCID1	The first ICCID	
ICCID2	The second ICCID	
ICCID3	The third ICCID	
...	The range of ICCIDs. We recommend you search for a maximum of 10 ICCIDs in one request to reduce the load on the server.	

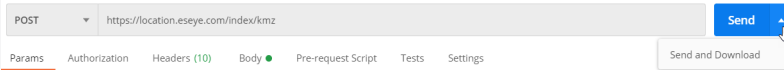
Example

Body raw JSON

```
{
  "request_id" : "1" ,
  "username" : "ExampleUsername" ,
  "password" : "ExamplePassword" ,
  "group" : "ExampleGroup",
```

```
"iccid" : [
  "8944538523000000000",
  "8944538523000000001",
  "8944538523000000002"
]
```

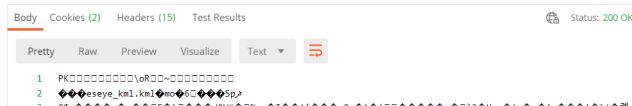
Using Postman or similar, we recommend you test the request using **Send** before you **Send and Download** the file.



Responses application/json

Test request using **Send**

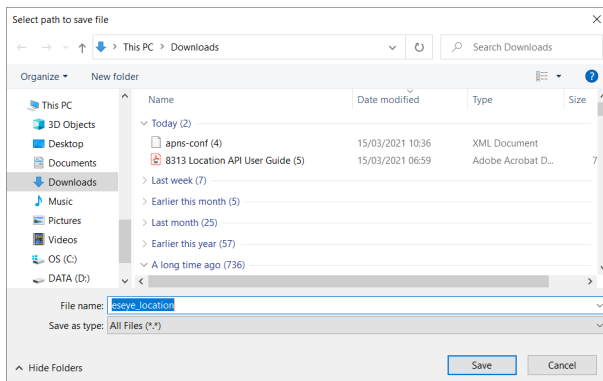
If you test the request using **Send**, then a successful response is similar to the following:



For information about errors, see *Troubleshooting errors* on page 15.

Download file using **Send and Download**

If you download the location map file using **Send and Download**, then a successful response immediately opens a dialog box to save the file to your default *Downloads* folder:



1. Add a `.kmz` extension to the **File name**.
2. Browse to where you want to save the file, then select **Save**.
3. Import the file into *Google Earth* or *Google My Maps* to view the SIM locations.



For more information, see the *Google Earth* or *Google My Maps* documentation.

Troubleshooting errors

When an error occurs, an error code and message are generated in the returned result:

```
{
  "status": {
    "status": "Error",
    "errorCode": "<ErrorCodeNumber>",
    "errorMessage": "<ErrorCodeMessage>"
  },
}
```

The following table can help you diagnose the error:

Error Code and Message	Reason for error
0 – Decoding failed: Syntax error	The JSON text is structured incorrectly. Ensure the JSON text follows the format described.  You may receive this error if you copied and pasted text from Microsoft Word. Some MS Word characters are slightly different from ASCII text. To avoid this, either type the text directly into Postman, or configure Microsoft Word to use basic ASCII text.
1 – The number of cells is 0	The mast, power, or ta values are absent or entered incorrectly.
601 – Missing parameter ICCID/MSISDN	The ICCID or MSISDN is absent or entered incorrectly, or the "s" in the https section of the request URL was not entered.
602 – Unauthorised Access, Invalid Account, or Invalid Password	This may indicate: - The username or password is incorrect. - The account does not exist in SIAM. Speak to your Account Manager for more information. - The account is not authorised to use the Location API. Speak to your Account Manager for more information.
603 – Parameter <parameter> value missing where <parameter> is the missing parameter	The reported parameter field name or value is absent or entered incorrectly. If the parameter is one of two parameters that you can use in the request, then the error refers to either parameter. For example, for <code>locatecalculate</code>, Parameter cellid value missing may also refer to the <code>ci</code> parameter.
605 – No msisdn found	The ICCID or MSISDN is absent or entered incorrectly, or the "s" in the https section of the request URL was not entered.
611 – invalid	The ICCID or MSISDN is entered incorrectly, or not available from this account.

Error Code and Message	Reason for error
<!DOCTYPE html>	Error message begins as follows: <pre><!DOCTYPE html> ...</pre> Indicates that one or more of the following sections in the Request URL is incorrect: - location - index/<request> where <request> is the request command, for example, <code>locatedevice</code>
Could not get any response	Error connecting to the server, for any of the following reasons: - You are not connected to the internet. - The Request URL is incorrect in one or more of the following sections: <pre>http :// .eseye.com</pre> - The server backend is not working properly. - SSL connections were blocked: fix this by importing SSL certificates in Chrome. - Cookies were not sent: fix this by using the Postman Interceptor extension. - Restricted headers were not sent: fix this by using the Postman Interceptor extension. - Request timeout: fix this by changing request timeout settings.



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